Randall C. Crawford

2019 86th Ct E – Inver Grove Heights, MN 55077
Cell: **(502) 243-5029** E-mail: rcc_0149@yahoo.com
LinkedIn Profile: https://www.linkedin.com/in/randall-crawford-43878875
Public GitHub Account: https://github.com/RCC0149/
LLC: Sweet Spot Analytics E-mail: support@sweetspotanalysis.com Website: www.sweetspotanalysis.com

Objective

Have a career fascination with the qualification, monitoring, and optimization of manufacturing operations, processes, and components. Believe in a data-driven and controlled organization. Diligently worked to develop a skill set that would ensure the accuracy and reliability of data collection, the efficiency and security of data transfer/processing and storage, the accessibility and usefulness of data reporting, and now robust data modeling and analysis. My current employment interest relates to process development, qualification, monitoring, and optimization, utilizing proven APQP and Six Sigma methodologies, strategic problem-solving methods, various metrology and prevention/detection tools, and data analysis/modeling techniques. Confident that the right opportunity will foster an industrious and innovative relationship that will lead to an extended, mutually beneficial partnership.

Education

MS in Data Science (Completed) **National University (online)** **10/24 ~12/25**

AI/Optimization Specialization Coursework Completed:

Graduate Capstone Project: Hybrid Graph Neural Network and Physics-Informed Neural Network for Predicting Interface Dynamics in Semiconductor Thin Film Deposition (ASE, GNN, PINN, MLP, Optuna)
Machine Learning Deployment (Git/GitHub, Flask, Heroku, Docker, Kubernetes, AWS)
Neural Network & Deep Learning (ANN, CNN, RNN, LSTM, GAN)
Applied Optimization Methods (Gradient, Convex, Random Walk, Markov Chain, Reinforcement Learning, Multi-objective, Genetic, Simulated Annealing, Ant Colony, Particle Swarm)

Additional Modeling Methods Covered and/or Utilized:

Supervised Learning Regression (Multiple Linear, Lasso, Elastic Net, Support Vector, Decision Tree, Random Forest, Gradient Boosting, XGBoost)
Supervised Learning Classification (Logistic Regression, Linear Discriminate Analysis, Decision Tree, Random Forest, XGBoost, Support Vector Machines, k-Nearest Neighbors, AdaBoost)
Unsupervised Learning Clustering (K-Means Hierarchical, Gaussian Mixture Models, Spectral)
Unsupervised Dimensionality Reduction (Principal Component Analysis, Linear Discriminate Analysis, Factor Analysis)
Ensemble Bagging (Random Forest, Decision Trees)
Ensemble Boosting (AdaBoost, Gradient Boosting, XGBoost)
Feature Selection (Backward Elimination, L1 Lasso, SelectKBest, Recursive Feature Elimination, Mutual Information, SHAP, Boruta)

Exploratory Data Analysis Steps Practiced:

Acquisition, Structure Determination, Filtration, Statistical Analysis, Visualizations, Data Modification/Transformation, & Model Refinement

Applications Primarily Utilized:

Python, R Studio, SAS Studio

Data Analytics Bootcamp (Completed) **Butler University (online)** **12/20 ~ 05/21**

Applications Covered:

Microsoft Excel & VBA Programming	*Python Programming*
Pandas Data Munging & Data Parsing	*Matplotlib Graphing*
Web Data (API Requests)	*JSON Traversal & Data Parsing*
SQL Queries & Modeling (PostgreSQL)	*Web Design (HTML / CSS)*
Web Frameworks & Dashboards (Bootstrap)	*MongoDB*
Flask Data Rendering	*ETL Project Work*
JavaScript Programming	*JavaScript APIs & Web Charting*
D3.js Fundamentals & Concepts	*Geomapping with Leaflet.js*
Data Wrangling with R	*Tableau/PowerBI Dashboarding*
Machine, Neural, & Deep Learning	*Big Data & Cloud ETL*

BA (Completed) **Indiana University Southeast, New Albany IN** **12/14**

- ***Bachelor degree in General Studies with concentration in Mathematics and Science***

Studies: *Six Sigma, TQM, SPC, & DOE*
TOC, MRP, MRP II, ERP, & JIT
Team Development & Problem Solving
Project Management & Lean Concepts
Industrial Engineering Solutions
Differential Equations & Applications
SQL Programming & VBA Modeling (Excel)
Propositional, Quantificational, & Symbolic Logic
Matrices, Graphs, Trees, & Machines
Regression & Variance Analysis (Excel & Minitab)
Discrete & Continuous Probability Distributions
Linear Programming & Modeling (Excel)

Relevant Work History

Contract Quality Systems Specialist **Boeing Corporation** **04/22 ~ 06/24**

Boeing is an American multinational corporation that designs, manufactures, and sells airplanes, rotorcraft, rockets, satellites, telecommunications equipment, and missiles worldwide.

Contractor: Mindlance Two 12-month remote contracts completed, and an additional 3-month remote contract to be completed by the end of June 2024.

Achievements

Supporting BCA Supplier Quality Core Technical (approx. 12 months):

- Provided a previously tested Excel GR&R study format for Supplier Quality use.
- Provided a previously tested Excel Capability Analysis study format for Supplier Quality use.
- Proposed ideas for the creation of supplemental guidance for PPAP MSA requirement (Application and Targeting for Common, Unique, and Key Characteristics).
- Managed a development and testing project for the use of the MBDvidia application from Capvidia to create FAIR forms and 2D/3D balloon drawings from 3D CATIA part models, 2D .pdf part drawings, and SSP documents, and upload them to BCA internal and external Net-Inspect supplier accounts to be completed by those suppliers.
    - Acquired waiver for the use of the application.
    - Worked with BCA managers to determine four project team members.
    - Located Capvidia Floating License server, and install necessary license key file for two floating licenses.
    - Used one of two training credits to have team members remotely trained on application.
    - Knowledgebase for 2D balloon drawing creation and 3D MBD tutorials were shared with team members.
    - Acquired Release of Boeing Proprietary Information (RBPI) approvals for 3D CATIA part models, 2D .pdf part drawings, and SSP documents to share with Capvidia for the development of QIF plug ins and translations.
    - Acquired Net-Inspect access key for team members to use to upload FAIR forms and 2D/3D balloon drawings.
    - Began pursuit of a Boeing endorsement for a collaboration between 3DS/Dassault and Capvidia to allow Capvidia to develop a QIF Plug in for their CATIA V5 models with Boeing SMEs.
    - Began the pursuit of a Step 242 translation of the RBPI approved CATIA V5 models to see, if it would improve translation capabilities.
- Managed a project to perform a security evaluation for the external use of the Axion Ray cloud application for Proof of Concept (PoC) projects within Boeing.
    - Worked with the assigned Boeing product owner to perform the required application vulnerability assessment.
    - A Fitness for Use (FFU) evaluation was completed.
        - Worked with project team and vendor to complete cloud and security documents, and had them reviewed by Enterprise Architecture before submittal.
        - Worked with vendor to develop a threat model for the cloud-based application, and worked with Information Security to identify and implement necessary mitigations.
        - Worked with purchasing agent (PA), project team members, and Strategic Sourcing & Contracts, to assure necessary contractual language was being included.

- Served as a Project Manager (PM) for the APQP team.
    - Facilitated a weekly APQP action items meeting.
        - Help support team member action items.
            - Guidelines for supplier PPAP artifact completion
            - Checklists for supplier PPAP artifacts
            - Supplier APQP work movement project process flows
    - Facilitated a weekly APQP pathfinder project meeting.
        - Help support team member projects.
            - Review of PFD and PFMEA submittals.
            - Review of MSA submittals.
    - Provided customer support for two active supplier APQP pathfinder projects.
            - Review of PFD, PFMEA, and Control Plan submittals.
            - Provided MSA recommendations.
- Develop an PPAP Interim Approval strategy that could be used with supplier work movement projects that were not afforded the required timeframe for a traditional APQP PPAP sequence to be rolled out.
- Worked as a Net-Inspect focal for BCA Supplier Quality.
    - Through evaluating the process of pulling APIs from Net-Inspect, Information Security and I were able to show the data team member the right Python module to perform the task.
    - Worked with Export Controls to improve the Document Acknowledgement prompt language in BCA Net-Inspect accounts to help guard against restricted uploads.
    - Worked with GPO to ensure that Net-Inspect was providing a link to Boeing's Privacy Policy to its users.
    - Provided clarity on Net-Inspect security and user levels on numerous occasions.
    - Fielded multiple inquiries about Net-Inspect APQP and FAI module capabilities.
    - Assisted Net-Inspect in their development of PFD, PFMEA, and Control Plan templates.
        - Would have helped them with the development of MSA and IPCS templates, if Net-Inspect would allow Boeing use of the aforementioned templates through APQP module.

<u>Supporting Enterprise Total Quality (approx. 15 months)</u>**:**

- Supported the utilization and evaluation of the ATS CM4D reporting application.
    - Utilized previously generated .csv file to create reporting for the 777X HBL drilling operation.
        - Used CM4D DataSmith to extract, translate, and parse data from the .csv file to be used by CM4D Classic.
        - Used CM4D Classic to create a variety of charting and reporting representations to be displayed in CM4D Interactive.
            - Evaluated statistical calculations and charting functionality with Minitab, and suggested improvements to ATS.
        - Used CM4D Interactive to set up filtering and charting representations to be queried and viewed by process engineers.
        - Worked with database support to develop a strategy using SQL extraction for the automatic generation of a .csv file for reporting applications.
    - Support Process Control Monitoring (PCM) team effort.
        - Provided assistance to other team member projects with problem solving insight regarding the CM4D DataSmith, Classic, Interactive, Batch Manager, Data Utility, and Scheduling Manager.
        - Worked with ATS support on problem solving related to team member projects.
        - Worked with an intern to develop a PCM project inquiry submittal interface questionnaire in R Studio, as well as PCM group mailbox to receive inquiries.
        - Created process flow for PCM project assessment, determination, and implementation, regarding measurement/detection control, data management, and PCM software reporting.
        - Worked with Boeing CM4D server support to install additional software, as well as diagnose and resolve server performance and data transfer issues.
- Assisted team member with the development of an APQP lesson learned submittal strategy.
- Worked with Enterprise and BCA Quality personnel to investigate the potential use of Alfresco for the storage of internal APQP project information and PPAP artifacts.
- Assisted training team members with the development of supplier APQP training.

- Worked with an assigned product owner for the Net-Inspect cloud application to perform a security evaluation to allow for the continued use of the FAI module and implementation of the APQP module by the Boeing enterprise and their suppliers.
    - An application vulnerability assessment was performed.
    - A waiver was obtained.
    - A Fitness for Use (FFU) evaluation was completed.
        - Worked with product owner and vendor to complete cloud and security documents.
        - Worked with vendor to develop a threat model for the cloud-based application, and worked with Information Security to identify and implement necessary mitigations.
        - Worked with purchasing agent (PA) and Enterprise Architecture to create required cost/benefit analysis.
        - Worked with purchasing agent (PA), Strategic Sourcing & Contracts, as well as Boeing and vendor legal representation to finalize contract language.
        - Worked with the Global Privacy Office (GPO) to insure the proper handling of user and admin personal information.
- Participated on a BDS Weapons team to initiate a trial project for use of the Net-Inspect FAI module.
    - Worked with team to define trial project objectives.
    - Provided testing metrics and strategy for the trial project.
    - Continued to assist team through FAI module inquiries with vendor and contract concerns.
- Participated on a BCA team to evaluate the Net-Inspect APQP module.
    - Worked with team to define desired capabilities for the APQP module.
    - Submitted those capabilities to Net-Inspect to confirm the current status of them.
    - Created an internal and external project simulation strategy to evaluate the capabilities and confirm the Net-Inspect status assessment.
    - Participated on a team to complete an internal and external project simulation for the Net-Inspect APQP module.
- Assisted team members with the development of an APQP prioritization tool related to models, commodities, assemblies, and components.
- Supported the utilization and evaluation of QDM reporting application.
    - Worked with SQL scripting support generate an appropriate .csv file to use with the application for the 777X HBL drilling operation.
    - Evaluated applications reporting functionality and calculations with Minitab, and suggested improvements to QDM.
- Coached and lead an APQP project for a BDS Tube Bending operation located near St. Louis, MO.
    - Worked with a cross-functional team (CFT) to create a comprehensive process flow diagram (PFD) and process failure mode effects analysis (PFMEA).
    - Various risks were identified.
    - Appropriate mitigation steps were proposed.

Training

- APQP Fundamentals	Remote	5 days
- Boeing Problem Solving	Remote	1 day

Travel

- APQP Tube Bending Project	St. Louis, MO	2 days (6 times)
- 2023 APQP Strategic Planning	Seattle, WA	5 days
- 2023 & 2024 Quarterly BCA SQ Site Visit	Everett, WA	5 days (3 times)

Contract Quality Engineer **Framebridge, Richmond KY** **07/21 ~ 01/22**

Framebridge is a manufacturer of wooden frames displaying various art, prints, and collectibles.
Contractor: RANDSTAD 6-Month Contract Completed

Achievements

- Utilized the 8-D problem solving method and video technology to resolve critical errors through corrective action.
- Performed numerous audits and assessments of processes, personnel, and components.

Quality Assurance Manager **Alliant Technologies LLC, Louisville KY** **10/16 ~ 02/20**

Alliant Technologies LLC is a provider of controls solutions, conveyor/sorting system designs, and custom manufactured PLCs to the freight and parcel, warehouse and distribution, manufacturing, airline, and government markets.

Achievements

- Conducted a comprehensive investigation into failures associated with an international airport project. Discovered a number of issues that were addressed formally. Best practices and company policies were created to prevent their reoccurrence.
- Assumed responsibility for the creation of check sheets from system drawings to be used by contracted electricians and company control engineers on UPS worksites for various levels of system commissioning. Inspected the system drawings for errors, and provided the design department with the necessary feedback to resolve them.
- Assisted in the development of policy for worksite communication and conduct.
- Suggested using data pads with construction monitoring software to create worksite daily reports efficiently, and improve timely reporting of issues to company and worksite managers.
    - Raken Jobsite Management Software
- Assisted the Panel Shop manager with the development of quality documentation and processes, then trained Panel Shop supervisors and wireman concerning them.
- Collected and stored quality data concerning panel manufacturing, then posted and reported results weekly to the Panel Shop manager and supervisors.
- Outlined a way to use QC data pads with data collection software to automatically generate, print, and store company and customer quality documentation to greatly reduce the labor, materials, and equipment depreciation involved in creating, copying, and scanning them.
- Followed up on customer corrective action documentation to make sure quality issues were being resolved in a timely manner.
- Logged in crimpers, electrical torque screwdrivers, calipers, and torque wrenches in a database for verification and calibration. Performed intermittent verification on those tools to determine their need for calibration.
    - GAGEPack Calibration Management Software
- Conceived of a FMEA strategy focusing on manufactured PLC defects.
- Put together a formal comparison of QMS and ISO standards.
- Systematically created a company safety manual that met third party audit standards with a score of 97%, qualifying the company for continued work with Amazon, GE, and others.
- Worked with HR and department managers to get required safety training and documentation in place.
- Suggested the use of video recording and screen recording for training and reference purposes, given most of the experienced employees were constantly on a worksite, and business expansion demanded the need for more employees.
- Suggested the need for regularly updated platform for project management that could be accessed by department managers and support groups to improve scheduling and prioritization of labor and activities.
- Audited project management documentation over various project types with a PMAT format. Provided project managers with feedback. Determined that specific documentation standards should be created for each project type.

Training

- 2018 CED Automation Exposition Seminars	Louisville, KY	1 day
- 2019 CED Automation Exposition Seminars	Louisville, KY	1 day

Travel

- Amazon System Commissioning	Buffalo, NY	5 days
- UPS Centennial Quality Response	Louisville, KY	5 days

    - Created a procedure for the verification, then managed a team to evaluate PLCs for a potential disconnect handle fault, and fix it.

- UPS System Commissioning	Phoenix, AZ	2 days

Qualification Technologist **Andersen Corporation, Bayport MN** **07/13 ~ 08/16**
Andersen Corporation is a manufacturer of doors and windows for the residential and commercial real estate development industries.

Contractor: ENTEGEE Engineering Technical Group **02/10 ~ 06/13**
Initial 9-Month Contract Completed **10/08 ~ 06/09**

Achievements

- The programming and operation of coordinate measuring equipment to generate dimensional measurement data for PPAP qualification activities (FAI layouts, GR&R studies, and capability analysis), product development projects, statistical process control, corrective action measures, and design drawing revisions (reverse engineering)
    - OGP Flash 200, ZIP 250, and FOV SmartScopes
        - Measure X, Measuremind NT, and MeasureFit Software Applications
            - Evaluated measuring fixtures, vinyl/fibrex extruded and milled wood profiles, profile gages, miter and bevel end cut angle fabrications, calibrator blocks, glue dies, aluminum profiles, capping and adhesive thicknesses, and purchased foam products
    - Romer AbsoluteArm 230
        - PC-DMIS For Portable 2014 Software Application
            - Evaluated curved wood and glass components, end cut fabrications, forming templates and molds, and measuring fixtures
    - Brown And Sharpe XCEL 7.10.7 Bridge CMM
        - PC-DMIS CAD++ 2013 Software Application
            - Evaluated extrusion dies, calibrator blocks, saw attachment designs, process standards, profile taper gauges, and measuring fixtures
- The design, verification, testing, and implementation of profile measuring fixtures for SmartScopes
- The development of SmartScope quality procedures concerning machine restart and re-alignment, profile sample preparation, fixture utilization, sample measurement, and sample retention
- The design, development, and revision of SmartScope dimensional profile roadmaps
    - Autocad Inventor
- The monthly SmartScope configuration data backup, calibration verification, and preventative maintenance check for five measuring machines
- The utilization and programming of data management software applications for automatic real time analysis, report generation, and server distribution of dimensional, testing, and process capability results from three SmartScope quality laboratories
    - QC-CALC Real Time and QC-CALC SPC Software Applications
        - Installed on each OGP SmartScope PC
    - SPC Office Buddy Software Application and Windows Batch File Programming
        - Installed on a designated data processing PC
- The weekly update of a SmartScope profile measurement data performance metric, and participation in a corrective action forum
- The completion of a $50k capital expenditure project to upgrade PCs, operating systems, measurement and data software applications on four OGP SmartScopes and a Brown and Sharpe CMM
- The use of various measuring equipment (Optical Comparator, Heidenhain XYZ Tables, Heidenhain Horizontal and Vertical Linear Scales, Caliper, Micrometer, Gage Pins and Blocks, Height Gage, Taper Gage, Level, Angle, etc…)
- The development and utilization of recognized company standard study formats for process capability and gage repeat and reproducibility (GR&R) using Microsoft Excel
- The utilization of a Minitab statistical analysis software for engineering problem resolution
- The performance of a Linearity study to determine the reliability of the DT500 Distance Sensor Laser
- The design and implementation of a QC workstation for the dimensional evaluation of curved wood blank components
    - Motorola LI4278 Barcode Scanner, Mitutoyu U-Wave System, and DT500 Distance Sensor
        - QC-GAGE and QC-CALC Real Time Software Applications
- The development and utilization of a mathematical shrinkage model to predict dimensional movement in curved wood blank components over time
- The use of multiple linear regression and correlation analysis to develop a predictive model to estimate the effects of a profile shaping process on curved wood blank components

- The development of an automatic time clock notification process for required control plan checks
    - Hourglass Time Clock Application and Windows Batch File Programming
- The training of employees concerning PPAP qualification standards, MSA, measurement software programming, GD&T, fixture design concepts, and general measurement strategies and techniques

Training (On-Site) (Productivity Quality, Inc.)

- MeasureFit Programming	Plymouth, MN	1 day	04/28/14

Training (Hexagon Metrology University)

- PC-DMIS For CMM 100-103	Plymouth, MN	5 days	02/14/14
- PC-DMIS For CMM 200-203	Plymouth, MN	5 days	03/27/15

CMM Programmer **Metrology Support Services, Fortville IN** **07/08 ~ 09/08**

Metrology Support Services is a provider of coordinate measuring equipment, retrofitting and calibration services, as well as inspection services for a wide variety of industries and applications.

Achievements

- The programming and operation of coordinate measuring equipment to generate dimension data and presentations for various clients
    - 2 Bridge CMMs: IMS Impact and MSS SX
        - VDMIS Version 6.2 Beta Software Application
- The development of a VBScript project for Microsoft Word to customize quote generation documentation derived from .html output from the company website

---

Senior Level CMM Programmer **Entegris Incorporated, Chaska MN** **11/07 ~ 05/08**

Entegris is a global supplier of technology, product, and service solutions that purify, protect, and transport critical materials used in the semiconductor and other high technology industries.
Contractor: ENTEGEE Engineering Technical Group *5-Month Contract Completed*

Achievements

- The systematic training of 5 technicians, 6 inspectors, and 2 auditors, in the effective use of coordinate measuring equipment and the PC DMIS dimensional measurement software application
- The programming and operation of coordinate measuring equipment to generate data for the verification or modification of assemblies, components, and gaging fixtures
    - 4 Bridge CMMs: Browne and Sharpe Global Classic and FX, Zeiss Eclipse and Numerex
        - PC DMIS CAD++ Version 4.2 MR1, 3.7 MR3, & 3.5 MR2 Software Applications
- The design of single and multi-shot CMM holding fixtures to improve measuring repeatability
- The design and implementation of effective datum structures to improve measuring repeatability, product transferability between CMMs, and standardization of setup methodology
- The design of a gold standard fixture, based on the Zeiss geometrical verification concept
- The development of interim ball bar and ring gauge CMM calibration programs and procedures, to verify the calibrated accuracy of CMMs
- The participation in the development of preventative maintenance procedures for each CMM

CMM Inspector **Dresser-Rand Corporation, Painted Post NY** **09/07 ~ 10/07**

Dresser-Rand is a global supplier of reciprocating compressors for the oil, natural gas, chemical, and naval industries. *Contractor: AAA Airporter Services, Inc. (MADI)* *45-Day Contract Completed*

Achievements

- The programming and operation of coordinate measuring equipment to generate data for the verification of machining processes and the modification of components

- DEA Delta SP 12' Manual Bridge CMM
    - Browne and Sharpe Tudor for Windows Software Application
- Sheffield Manual Portage 76-B Rotary Table CMM
    - ATT Inc. CAPPS DMIS Version 6.1 Software Application

- The restoration of the Sheffield CMM, consequently increasing lab measurement throughput 80%
- The development of qualification and preventative maintenance procedures for the Sheffield CMM

Contract QE Assistant **Medventure, Louisville KY** **11/06 ~ 01/07**
Medventure is a supplier of outpatient surgical devices for the healthcare industry. *Contract Completed*

Achievements
- The development of completed supplier qualification reports for all of the company's vendors.

CMM Programmer **Montaplast of North America, Frankfort KY** **01/05 ~ 01/06**
Montaplast is a global supplier of welded lost core and injection molded plastic parts for the automotive industry.

Achievements
- The programming and operation of coordinate measuring equipment to generate measurement data for the buy-off (PPAP), certification, development, liability, and/or modification of assemblies, components, gaging fixtures, primary and secondary standards, and molds
    - Browne and Sharpe XCEL Single Column CMM
        - QUINDOS 6.0, CAM2 AUTO, and PC DMIS 3.5 Software Applications
- The familiarity with reading German and Japanese blueprints
- The use of various measuring equipment (QL-20 Contour Projector Optical Comparator, Mahr Perthometer, Caliper, Micrometer, Gage Pins and Blocks, Height Gage, Taper Gage, etc…)
- The troubleshooting of VBScript in data analysis spreadsheets

Training (On-Site)

- QUINDOS 6.0 Programming	10 days	QUALIS (Instructor)
- Descriptive Statistics & DOE	5 days	
- Geometrical Dimensioning & Tolerancing	5 days	

CMM Layout Technician **Oxford Automotive, Corydon IN** **06/98 ~ 03/01**
Oxford Automotive Products Company was a global supplier of stamped and/or welded metal chassis parts for General Motors, Ford, and Daimler-Chrysler

Achievements
- The maintenance, operation, and programming of coordinate measuring equipment to generate presentations for the buy-off(PPAP), certification, development, liability, and/or modification of parts, gaging fixtures, primary and secondary standards, tools, and dies
    - Trimek LK LY90S Twin Column CMM
        - LK DMIS 5.3 and LK CAMIO 5.3 Software Applications
    - Mora Manual Single Column CMM
        - GEOMET, upgraded to Metronics QC5000 Software Application
- The detailed outline of operating and programming instructions for the automated CMM
- The development of data collection procedures and forms, statistical analysis spreadsheets, and graphs for Attribute R & R, Bias, Capability, Gage R & R, Linearity, SPC, and Stability studies
- The formation of calibration procedures, the maintenance of a calibration database, and the generation of calibration work orders for all utilized measuring tools
    - GageTrack Calibration Software
- The practical familiarity with GD&T for Ford and GM parts, PPAP demands, and QS-9000 standards

- The conception of a comprehensive historical reference for quality dilemma (pertinent details, visual renditions, corrective action recommendations) and an inter-departmental system for its maintenance

Training

- Measurement Systems Analysis Certificate	Indianapolis, IN	1 day	07/09/99
- LK DMIS Programming Certificate	Brighton, MI	5 days	05/26/00
- LK CAMIO CAD Interface Certificate	Brighton, MI	5 days	06/09/00

Travel

- Oxford PPAP of GM parts — Ramos Arizpe, Mexico — 24 days — 11/01/99 ~ 11/24/99
  4 days — 11/30/99 ~ 12/03/99
  - CMM operation for part layout requirements, graph generation for capability study requirements (StatGraphics), and spreadsheet development for Trim Analysis requirements
- Gaging Fixture Buy-offs — Yokohama, Japan — 13 days — 06/12/00 ~ 06/23/00
  >> Sugai Auto Works
  - Evaluated gaging fixture quality and design, created and demonstrated a Gage Repeat and Reproducibility Study for Japanese use, and developed a checklist for gaging fixture buy-offs
- Oxford PPAP of Ford parts — Oscoda, MI — 4 days — 07/05/00 ~ 07/08/00
  - Demonstrated layout methods for gaging fixtures and parts without the use of a CMM
- Assembler/Tool Buy-offs — Grand Rapids, MI — 2 days — 07/11/00 ~ 07/12/00
  >> Advanced Tooling Systems
  - Evaluated weld assembly equipment
  >> Engineered Tooling Systems
  - Evaluated tooling equipment through part quality
- Piercing Unit Buy-off — Troy, MI — 2 days — 07/13/00 ~ 07/14/00
  >> Oxford Owned Piercing Unit Designer (Hillman Machinists)
  - Evaluated piercing equipment
- Part Quality Buy-offs — Yokohama, Japan — 41 days — 07/23/00 ~ 09/01/00
  >> Kanto Auto Works, Topre, Nikkei, Seki-tech, Katayama, Suzuki, and Sugai Auto Works
  - Evaluated part and gaging fixture quality, created a Capability Study format for Japanese use, and electronically documented part and fixture quality reports

---

Product Flow Analyst — **Golden Brands/Golden Foods, Louisville KY** — **05/97 ~ 04/98**
Three Promotions: Lab Technician > Warehouse Quality Inspector > Inventory Control Specialist

Golden Brands/Golden Foods Inc. was a supplier of shortening for Pillsbury and several regional restaurant franchises.

Achievements

- The routine interface with MIS, production planning, quality control, sales, and shipping, to develop more effective queries for the inventory database and acquire weekly production and shipping schedules to aid the task of scheduling daily product flow activities
- The scheduling of daily allocation, storage, and retrieval activities for finished product and materials
Averages: 13 trailers/day 23 pallets/trailer 32 boxes/pallet 8 customers/day 2 loaders 2 line drivers
Factors: Products-10 types Boxes-4 types Loaded Pallets-4 types Cooling-3 rooms & 5 periods
Racking (single entry)-3 types & 4 depths (24,15,9,3 pallets) FIFO-maintained for each destination
- The creation of an algorithmic schematic for the pre-determination of scheduling activities
- The development and dispersal of a daily finished product and materials allocation, storage, and retrieval form to provide direction for line drivers and dock loaders
- The daily reconciliation of product flow activities through the comparison of completed forms, database activity, and physical inventory
- The quarterly inventory of approx. 7,000,000 lbs. of shortening at an accuracy of 99.97%
- The development of Quality procedures and data collection forms for the Shipping department
- The summation of individual, dock loader, line driver, shift, and collective performance measurement indicators (Man-hours vs. Product Moved vs. Product Reworked vs. Product Scraped)

- The successful comparison of quantitative data from Receiving, Processing, Production, Inventory, and Shipping to isolate the primary cause of waste in the total operation, and the successful development of procedures for the Production department to rectify it
- The utilization of projected sales, inventory fluctuations, and traffic study to estimate the extent, type, and location of additional racking
- The submission of conventional drawings for the floor layout of the racking expansion, and the supervision of the installation

Training

- Intermediate Microsoft Excel Course Louisville, KY 1 day 03/98 approx.

Travel

- Shipment Inspection Joplin, MO 2 days 01/98 approx.
  >> Carmar Freezers Inc.
  - Product temperature sample and trailer inspection

Quality Technician **Futura Plastics, Louisville KY** **02/97 ~ 04/97**
Agency Employee: Kelly Services, Inc. Louisville, KY *Temporary Assignment*

Futura Plastics, Inc. was a supplier of plastic parts for General Electric.

Achievements

- The acquaintance with the plastic injection molding process and quality issues common to it
- The pulling of an over-exposed supervisor from a nylon gas leak

---

Scale House Attendant **Consolidated Grain and Barge, Jeffersonville IN** **10/96 ~ 01/97**
Agency Employee: Personnel Management, Inc. Jeffersonville, IN *Seasonal Employment*

Consolidated Grain and Barge, Inc. is a distributor of grains, fertilizers, and road salt.

Achievements

- The tare and weighing of tractor-trailers and dump trucks for the generation of a billing statement for fertilizers (potash, ammonium nitrate, etc.) and road salt
- The sampling and testing (moisture, foreign material, etc.) of grain (corn, soybeans, wheat, etc.), and the weighing and tare of tractor-trailers and dump trucks for the generation of a credit voucher
- The proper generation and distribution of billing statements and credit vouchers

---

Manufacturing Associate **A. O. Smith Automotive, Corydon IN** **05/90 ~ 03/94**
A. O. Smith was purchased by Tower Automotive

A. O. Smith Automotive Products Company was a supplier of stamped and/or welded metal chassis parts for Ford Motor Company.

Achievements

- The performance of every functional position, barring support and management roles, at the facility
- The election to the position of Team Leader by fellow team members
- The experience of meeting facilitation, agenda setting, and group problem solving

- The conception of a Peer Review system for Associate empowerment in the areas of performance evaluation, continuous improvement, reward and recognition, corrective action, hiring, and termination
- The periodic summarization of assembly defects and their respective catalysts
- The re-allocation of labor and introduction of inspection responsibilities in a bottleneck work station such that the number of operators was reduced from 5 to 4 (providing an additional Associate for continuous improvement activities), the cycle time was reduced by approximately 4 seconds (moving the bottleneck to the next workstation, a series of robotic welding operations), the number of defects found after the re-organized work station was reduced by 75% (relieving the new bottleneck of poor quality input), and all of the Associates involved expressed their appreciation

Training (On Site)

- Dale Carnegie Interpersonal Training
- Myers-Briggs Type Indicator (E/I NTJ) And Problem Solving
- 8-D Problem Solving Seminar
- TQM Orientation
- Cost Justification
- Peer Review Process

Personal Interests

- Strategic Games (Titan, Chess, etc.), Music (Classical and Contemporary), and Philosophy (Axiology and Logic)

References

Available Upon Request